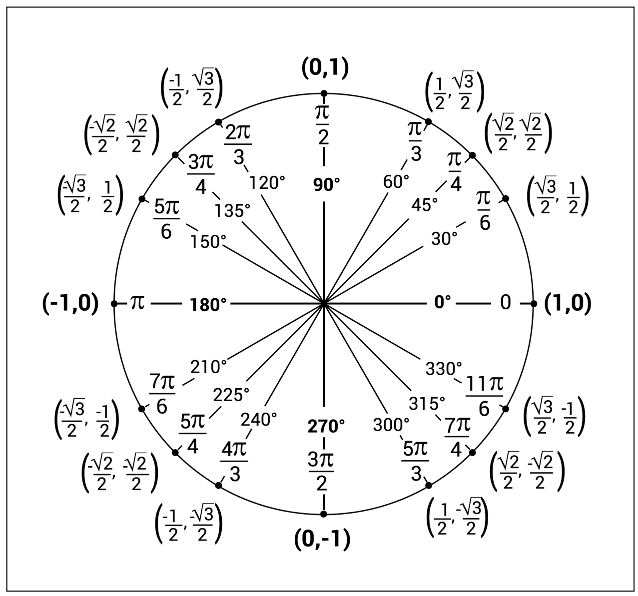
Math 125

Extra Unit A practice 2

The problems that follow are examples of questions I've asked on assessments in past semesters. They are meant to give you a sense of the style of problems on assessments, but not necessarily the content of the problems. For example, just because a particular concept doesn't appear in these practice problems doesn't mean that you aren't responsible for knowing it for our assessments. You're responsible for knowing anything in the notes, the past final worksheets, and the Brightspace problems. The formulas below will be provided on Opportunity A and Jubilee 1.

Formulas

- $\lim_{\theta \to 0} \frac{\sin\theta}{\theta} = 1$, $\lim_{\theta\to 0} \frac{\cos\theta - 1}{\theta} = 0$
- Continuity at $x = a$: $\lim_{x \to a} f(x) = f(a)$
- $f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$
- Basic derivatives
  - $\frac{d}{dx}(c \cdot f(x)) = c \cdot f'(x)$
  - $\frac{d}{dx}(f(x) \pm g(x)) = f'(x) \pm g'(x)$
  - $\frac{d}{dx}(x^n) = nx^{n-1}$
  - $\frac{d}{dx}(e^x) = e^x$
  - $\frac{d}{dx}(\ln(x)) = \frac{1}{x}$
  - $\frac{d}{dx}(\sin x) = \cos x$
  - $\frac{d}{dx}(\cos x) = -\sin x$
  - $\frac{d}{dx}(\tan x) = \sec^2 x$
  - $\frac{d}{dx}(\sec x) = \sec x \cdot \tan x$
  - $\frac{d}{dx}(\cot x) = -\csc^2 x$
  - $\frac{d}{dx}(\csc x) = -\csc x \cdot \cot x$
- Derivative rules
  - $\frac{d}{dx}(f(x) \cdot g(x)) = f'(x) \cdot g(x) + g'(x) \cdot f(x)$
  - $\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x) \cdot f'(x) - f(x) \cdot g'(x)}{[g(x)]^2}$
  - $\frac{d}{dx}(f(g(x))) = f'(g(x)) \cdot g'(x)$
- $L(x) = f'(a)(x-a) + f(a)$



A1 - Limits and continuity

(a) Let $f(x)$ and $g(x)$ be functions such that $-g(x) \leq f(x) \leq g(x)$ for all $x \neq 2$, and $g(2) = 0$. If $g(x)$ is continuous at $x = 2$, must the following limit necessarily exist?

$$\lim_{x \to 2} f(x)$$

Explain why or why not.

If $g(x)$ is continuous at $x=2$, then $\lim_{x\to 2} g(x) = g(2) = 0$, so

$$0 = \lim_{x\to 2} -g(x) \leq \lim_{x\to 2} f(x) \leq \lim_{x\to 2} g(x) = 0$$

$\rightarrow$ by the Squeeze Theorem $\lim_{x\to 2} f(x) = 0$.

(b) Find the limits. Clearly show each step of your solution for full credit. If the limit does not exist, explain why.

(i) $\lim_{x\to\infty} \dfrac{x + 1 - \sqrt{x^2 + x + 1}}{1} \cdot \dfrac{x+1+\sqrt{x^2+x+1}}{x+1+\sqrt{x^2+x+1}} = \lim_{x\to\infty} \dfrac{(x+1)^2 - (x^2+x+1)}{x+1+\sqrt{x^2+x+1}}$

$= \lim_{x\to\infty} \dfrac{x}{x+1+\sqrt{x^2+x+1}} = \lim_{x\to\infty} \dfrac{1}{1+\frac{1}{x}+\sqrt{1+\frac{1}{x}+\frac{1}{x^2}}} = \dfrac{1}{1+\sqrt{1}} = \boxed{\dfrac{1}{2}}$

(ii) $\lim_{x\to 3^-} \dfrac{x^2 - 9}{|x - 3|} = \lim_{x\to 3^-} \dfrac{(x-3)(x+3)}{-(x-3)} = \lim_{x\to 3^-} -(x+3) = \boxed{-6}$

$x < 3$

(c) Let $f(x) = \begin{cases} x^2 & \text{if } x \leq 1 \\ a + b(1 - \frac{1}{x}) & \text{if } x > 1 \end{cases}$

For what values of a and b is f continuous at $x = 1$?

$f(1) = \lim_{x\to 1^-} f(x) = 1^2 = 1$

$\lim_{x\to 1^+} a + b(1-\frac{1}{x}) = a + b(1-\frac{1}{1}) = a$

$a = 1$, b can be anything.

A2 - **The derivative**

(a) Evaluate the limit

$$\lim_{\theta \to \frac{\pi}{3}} \frac{\cos\theta - \frac{1}{2}}{\theta - \frac{\pi}{3}}$$

by finding a function $f(\theta)$ and a point a such that the given limit is equal to the derivative of $f(\theta)$ at a.

$$\lim_{\theta \to \frac{\pi}{3}} \frac{\cos\theta - \cos(\frac{\pi}{3})}{\theta - \frac{\pi}{3}} = f'(\tfrac{\pi}{3}) \quad \text{where } f(\theta) = \cos\theta$$

$$f'(\theta) = -\sin\theta$$

$$f'(\tfrac{\pi}{3}) = -\sin(\tfrac{\pi}{3}) = \boxed{-\frac{\sqrt{3}}{2}}$$

(b) Consider the function g that is given by

$$g(x) = \begin{cases} 3x & x < 1 \\ x^2 + 2x & x \geq 1 \end{cases}$$

Use the definition of the derivative at a point to determine if g is differentiable at $x = 1$.

$$\lim_{h\to 0^+} \frac{g(1+h) - g(1)}{h} = \lim_{h\to 0^+} \frac{(1+h)^2 + 2(1+h) - (1^2 + 2\cdot 1)}{h} = \lim_{h\to 0^+} \frac{h^2 + 4h}{h} = \lim_{h\to 0^+} h + 4 = 4$$

$$\lim_{h\to 0^-} \frac{g(1+h) - g(1)}{h} = \lim_{h\to 0^-} \frac{3(1+h) - (1^2 + 2\cdot 1)}{h} = \lim_{h\to 0^-} \frac{3h}{h} = 3$$

$\neq$

so $g'(1)$ DNE.

A3 - Derivative rules

(a) If

$$f(x) = 2\sin(x)\ln(2 - \ln(x) - x)$$

find $f'(x)$. You do not need to simplify your answer.

$$f'(x) = 2\cos x \cdot \ln(2 - \ln(x) - x) + 2\sin x \cdot \frac{1}{2 - \ln x - x} \cdot \left(-\frac{1}{x} - 1\right)$$

(b) If $h(x) = (-x+3)^{\frac{x+1}{x}}$, find $h'(x)$. You do not need to simplify your answer.

$$h(x) = e^{\frac{x+1}{x} \cdot \ln(-x+3)}$$

$$h'(x) = e^{\frac{x+1}{x} \cdot \ln(-x+3)} \cdot \left(\frac{x \cdot 1 - (x+1)}{x^2} \cdot \ln(-x+3) + \frac{x+1}{x} \cdot \frac{1}{-x+3} \cdot (-1) \right)$$

A4 - Implicit curves and linearization

(a) Find the slope of the curve

$$xe^{y^2+2y+1} = y\sin x + \frac{\pi e}{2} + 1$$

at $\left(\frac{\pi}{2}, -1\right)$.

$$1 \cdot e^{y^2+2y+1} + x \cdot e^{y^2+2y+1} \cdot (2y+2) \cdot \frac{dy}{dx} = y\cos x + \frac{dy}{dx}\sin x$$

@ $\left(\frac{\pi}{2}, -1\right)$: $1 \cdot e^{(-1)^2+2\cdot(-1)+1} + \frac{\pi}{2} \cdot e^{(-1)^2+2\cdot(-1)+1} \cdot (2(-1)+2) \cdot \frac{dy}{dx} = -1 \cdot \cos\left(\frac{\pi}{2}\right) + \frac{dy}{dx}\sin\left(\frac{\pi}{2}\right)$

$$1 \cdot 1 + \frac{\pi}{2} \cdot 1 \cdot 0 \cdot \frac{dy}{dx} = 0 + \frac{dy}{dx} \rightarrow \frac{dy}{dx} = 1$$

(b) Find the linearization of $f(x) = \sqrt[3]{1+3x}$ at $a = 0$, and use it to give an approximate value for $\sqrt[3]{1.03}$.

$f'(x) = \frac{1}{3}(1+3x)^{-\frac{2}{3}} \cdot 3$ $\qquad f(0) = 1$

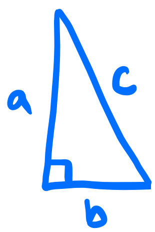
$f'(0) = 1$

$y = 1 \cdot (x-0) + 1$

$\sqrt[3]{1.03} = f(0.01) \approx 1 \cdot (0.01 - 0) + 1 = 1.01$

A5 - Related rates

(a) One leg of a right triangle is 5 feet and decreasing at a rate of 1 foot per second while the hypotenuse is 13 feet and increasing at a rate of 7 feet per second. What is the rate of change of the length of the other leg?



$$a^2+b^2=c^2$$

$$2a\cdot\frac{da}{dt}+2b\cdot\frac{db}{dt}=2c\cdot\frac{dc}{dt}$$

$$2\cdot 5\cdot -1+2\cdot 12\cdot\frac{db}{dt}=2\cdot 13\cdot 7$$

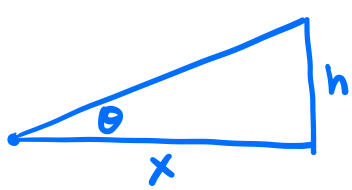
$$5^2+b^2=13^2$$

$$\rightarrow b=12$$

$$-10+24\cdot\frac{db}{dt}=182$$

$$24\cdot\frac{db}{dt}=192 \rightarrow \boxed{\frac{db}{dt}=8}$$

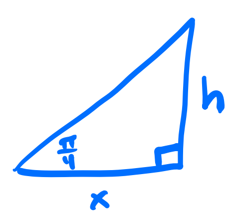
(b) Suppose you're sitting east of a tall building in the late afternoon watching the sun set over the building. At the moment the sun goes behind the building (from your perspective), the tip of the building's shadow passes you at a rate of 500 feet per hour. Also, the angle of elevation from where you sit to the tip of the building is $\frac{\pi}{4}$ radians. If the angle of elevation of the sun from your perspective is changing at -0.25 radians per hour, how far away are you sitting from the base of the building?



$\frac{dh}{dt}=0$ (building height is constant)

$$\frac{d\theta}{dt}=-0.25$$

$$\frac{dx}{dt}=500$$



$$\tan\theta=\frac{h}{x}$$

$$\sec^2\theta\cdot\frac{d\theta}{dt}=\frac{x\cdot\frac{dh}{dt}-h\cdot\frac{dx}{dt}}{x^2}$$

$$h=x\tan\left(\frac{\pi}{4}\right)=x$$

so $h=x$

$$\sec^2\left(\frac{\pi}{4}\right)=\frac{1}{\cos^2\left(\frac{\pi}{4}\right)}=\frac{1}{\frac{1}{2}}=2$$

$$2\cdot -0.25=\frac{x\cdot 0-x\cdot 500}{x^2}$$

$$-0.5=\frac{-500}{x} \rightarrow \boxed{x=1000}$$